Does spatial context transfer? Neighbourhood-aware cell typing in seminiferous tubules under tissue-organization and species shift

Anonymous Author(s)
Affiliation
Address
email

Abstract

Methods for annotating spatial transcriptomics data increasingly use a spot's tissue neighbourhood as input, but they are benchmarked within one tissue and condition, so it is unknown whether the spatial signal they learn holds when tissue organization changes. We build a small benchmark from the public Slide-seqV2 testis pucks of Chen et al. (2021): three wild-type mouse, three diabetic *ob/ob* mouse, and two human pucks (252,898 beads), with cell-type labels from NMFreg (non-negative matrix factorization regression). Using germ-cell stage annotation in seminiferous tubules as the task, we compare an expression-only classifier with neighbourhood-augmented features, a GraphSAGE model, and test-time smoothing of the expression-only posterior, under held-out-puck, wild-type-to-*ob/ob*, and mouse-to-human transfer. Spatial context adds about one macro-F1 point within a species and up to three on human pucks. The *ob/ob* pucks are not a distribution shift for this task. Across species, expression-only accuracy drops by ten points; neighbourhood-augmented features and test-time smoothing keep their gains, whereas GraphSAGE, which matches them within a species, becomes the worst model, losing five points and fifteen F1 points on round spermatids. The gain from context tracks how informative the neighbourhood is relative to the bead itself rather than the target tissue's neighbourhood homogeneity. Fixed spatial aggregation transfers across the shifts we tested where learned spatial features do not.

1 Introduction

Most recent methods for annotating cells or beads in spatial transcriptomics use the tissue neighbourhood of a spot as an additional input, either by concatenating a spot's own expression with an aggregate of its neighbours' [Singhal et al., 2024, Varrone et al., 2024] or by learning a graph neural network (GNN) over a spatial adjacency graph [Hu et al., 2021, Dong and Zhang, 2022, Long et al., 2023]. Benchmarks of these methods [Yuan et al., 2024, Li et al., 2022] evaluate within one tissue and one condition, so they do not tell us whether the spatial signal a model learns still holds when tissue organization changes, which is the question that matters when a model trained on a reference is used to annotate samples from other conditions, individuals, or species.

The testis is a natural place to ask it. Spermatogenesis proceeds inside seminiferous tubules along a radial axis, from spermatogonia at the basement membrane through spermatocytes and round spermatids to elongating spermatids at the lumen, and tubule cross-sections cycle through a fixed sequence of stages [Hess and de Franca, 2008, Chakravorty et al., 2026]. The same layering exists in mouse and human, but the human cycle differs in stage number and in how stages tile a cross-section. Chen et al. [2021] produced Slide-seqV2 [Stickels et al., 2021] pucks of wild-type mouse testis, of testis from leptin-deficient diabetic *ob/ob* mice, in which they report disrupted spatial organization of the tubules, and of normal human testis, each with bead-level cell-type labels from NMFreg [Rodriques et al., 2019]. This gives three levels of distribution shift on one tissue with one platform and one labelling procedure: a held-out puck of the same condition, a condition that alters tissue organization, and a different species.

We ask how much spatial context improves bead-level annotation of germ-cell stage, and whether the improvement survives each shift, comparing an expression-only classifier against neighbourhood-augmented features, a graph neural network, and test-time smoothing of the expression-only posterior over the target puck's spatial graph. The last is trained without any spatial input and uses the target tissue's geometry only at inference, separating the value of a spatial prior from the transferability of learned spatial features.

2 Data and task

Pucks. We use the eight processed Slide-seqV2 pucks released by Chen et al. [2021]: three wild-type (WT) mouse, three *ob/ob* diabetic (DB) mouse, and two normal human (HU) pucks, each with a bead-by-gene count matrix, bead coordinates, and a per-bead label over nine types (spermatogonia SPG, spermatocytes SPC, round spermatids RS, elongating/elongated spermatids ES, Sertoli, myoid, Leydig, endothelial, macrophage) assigned by NMFreg, a non-negative matrix factorization regression against a single-cell reference. We keep beads with a label and at least 100 unique molecular identifiers (UMIs), leaving 25,377 to 39,181 beads per puck (252,898 in total; Table 2).

Cross-species gene space. All pucks are restricted to one-to-one mouse–human orthologs from Mouse Genome Informatics (MGI) [Baldarelli et al., 2024] detected in every puck (13,573 genes), so a model trained on mouse can be applied to human without adaptation. Counts are library-size normalized and log-transformed [Wolf et al., 2018]. We select 2,000 highly variable genes on the WT pucks only [Stuart et al., 2019], then standardize each gene within each puck using that puck's own mean and standard deviation. This unsupervised per-puck standardization is the only batch handling we apply.

Task. The primary task is four-class germ-cell stage annotation (SPG, SPC, RS, ES) on germ-cell beads, which are 81 to 90% of beads. This is the layer the radial tubule axis defines, and the reproducible part of the NMFreg annotation: within-puck and cross-puck macro-F1 agree (0.71 vs. 0.70), whereas the rare somatic types (endothelial, macrophage, myoid) are near chance from a single bead's expression under any split. Six-class and nine-class variants are in the appendix.

Shift settings. Training pucks are always disjoint from the test puck. *WT→WT*: train on two WT pucks, test on the third (three folds). *WT→ob/ob*: train on all three WT pucks, test on each DB puck. *WT→human*: train on all three WT pucks, test on each human puck; *mouse→human* adds the DB pucks to training. As in-distribution references for the shifted targets, *ob/ob→ob/ob* and *human→human* train on the other pucks of the same condition. The NMFreg labels were computed from expression alone, so a gain from spatial features cannot be an artefact of how the labels were made. For each puck we also compute neighbourhood homogeneity, the mean fraction of a bead's 15 nearest spatial neighbours that share its label; it is used only to describe the pucks, never as an input. Among germ-cell beads it is 0.53 to 0.56 on WT, 0.49 to 0.57 on *ob/ob*, and 0.38 to 0.40 on human pucks.

3 Models and evaluation protocol

All models operate on a 50-dimensional principal component analysis (PCA) embedding z_i of the standardized expression, with the PCA fit on the training pucks only and applied to the test puck. Let $\mathcal{N}_k(i)$ be the k nearest spatial neighbours of bead i within its puck, and $\bar{z}_i^{(k)} = \frac{1}{k}\sum_{j\in\mathcal{N}_k(i)} z_j$ the neighbourhood mean, computed over all beads of the puck (non-germ included), since context is label-free.

Table 1: Macro-F1 for germ-cell stage under six settings (left three in-distribution, right three shifted); mean $\pm$ standard deviation (SD) across test pucks. nbr: mean embedding of the 15 nearest spatial neighbours. α selected on the WT→WT folds. MLP and neighbourhood-only rows: Table 3.

Model	WT→WT	ob/ob→ob/ob	human→human	WT→ob/ob	WT→human	mouse→human
LR (expr)	0.699 ±0.013	0.717 ±0.022	0.486 ±0.008	0.713 ±0.026	0.385 ±0.005	0.385 ±0.006
LR (expr+nbr)	0.711 ±0.013	0.723 ±0.016	0.513 ±0.011	0.721 ±0.023	0.398 ±0.007	0.396 ±0.008
LR (expr) + smooth (α=0.5)	0.712 ±0.016	0.733 ±0.019	0.501 ±0.008	0.732 ±0.021	0.401 ±0.005	0.400 ±0.006
GraphSAGE	0.711 ±0.015	0.716 ±0.023	0.505 ±0.013	0.721 ±0.023	0.347 ±0.012	0.350 ±0.012

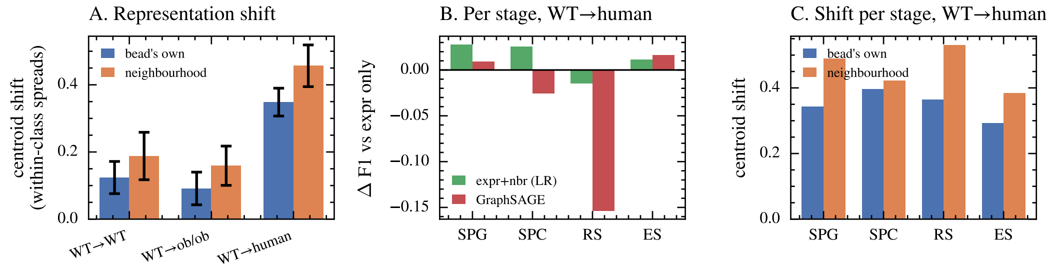

Figure 1: (A) Distance each germ stage's centroid moves from training to test puck, in units of its within-class spread in training, for the bead's own 50-dimensional embedding and for the neighbourhood mean; bars are means over stages and test pucks, error bars their standard deviation. (B) Per-stage F1 change relative to the expression-only LR, and (C) per-stage centroid shift, both under species shift. Round spermatids are where GraphSAGE loses most and where the neighbourhood representation moves furthest.

Expression only. A multinomial logistic regression (LR) on z_i, and a two-layer multilayer perceptron (MLP; 256 hidden units, dropout 0.2). Both use class-balanced loss weights.

Neighbourhood-augmented features. LR and MLP on $[z_i, \bar{z}_i^{(k)}]$, the construction used by BANKSY [Singhal et al., 2024] and related methods, for $k \in \{5, 15, 30\}$. As a diagnostic we also fit LR on $\bar{z}_i^{(k)}$ alone, i.e. on the surroundings without the bead.

Graph neural network. A two-layer GraphSAGE with mean aggregation [Hamilton et al., 2017], $h_i' = \sigma(W_1 h_i + W_2 \bar{h}_i^{(15)})$, 128 hidden units, trained full-batch on the union of training pucks (each puck is its own connected graph) and applied to the test puck's graph.

Test-time smoothing. We take the expression-only LR posterior p_i on the test puck and replace it with $p_i' = (1-\alpha)\, p_i + \alpha\, \bar{p}_i^{(15)}$, then predict $\arg\max p_i'$. This is one step of label propagation on the model's own outputs [Huang et al., 2021]. We sweep $\alpha \in \{0.25, 0.5, 0.75, 1\}$.

Evaluation. Macro-F1 over the classes present in the test puck and per-class F1, averaged over seeds (three for MLP and GraphSAGE; LR is deterministic) and then over the test pucks of a setting, with the standard deviation across test pucks. Hyperparameters were fixed a priori and shared across all settings, and nothing was tuned on any test puck; the values are listed in the appendix. The full grid runs on two CPU cores in a few hours.

4 Results

In-distribution gains are small and consistent. On held-out WT pucks the expression-only logistic regression reaches 0.699 macro-F1, and every form of spatial context adds about one point: 0.711 for neighbourhood-augmented LR and for GraphSAGE, 0.712 for test-time smoothing (Table 1). The gain is small but systematic: positive on all 15 setting–test-puck folds (Wilcoxon $p = 6 \times 10^{-5}$), with a bead-level paired bootstrap excluding zero on 13 of 15. A neighbourhood-only classifier reaches 0.636, so most of the stage signal is in the bead itself. Gains are similar on *ob/ob* and larger on human folds, where the expression-only ceiling is lower (0.486) and neighbourhood features add 2.7 points (0.513).

The *ob/ob* shift is small for this task. Models trained on WT pucks perform as well on *ob/ob* pucks as models trained on other *ob/ob* pucks (0.713 vs. 0.717 expression only; 0.721 vs. 0.723 with neighbourhood features). The spatial gains carry over unchanged. A classifier trained to tell WT from *ob/ob* beads inside the models' feature space performs at chance (area under the curve, AUC: 0.499 own, 0.519 neighbourhood; Table 8), so the condition is not a covariate shift in the space the models see, and the disruption reported by Chen et al. [2021] does not register at this resolution.

Species shift is large and is dominated by expression. Expression-only macro-F1 falls from 0.486 (human→human) to 0.385 (WT→human), and adding the *ob/ob* pucks to training does not change it (0.385). Neighbourhood-augmented LR retains a gain under species shift (0.398, $+1.3$), as does test-time smoothing (0.401, $+1.6$). GraphSAGE does not: it is within 0.008 of neighbourhood-augmented LR in every in-distribution setting and becomes the worst model across species, at 0.347 (-3.8 vs. expression only, -5.1 vs. neighbourhood LR; seed-to-seed SD ≤ 0.007). Per stage (Fig. 1B), the loss is concentrated on round spermatids, the largest human class, whose F1 drops from 0.390 to 0.237 with GraphSAGE and to 0.376 with neighbourhood LR; with neighbourhood LR, spermatogonia and spermatocytes still gain about 3 points. Depth is what costs it: one, two and three layers are indistinguishable in distribution (0.695 to 0.697) but fall monotonically across species (0.361, 0.348, 0.339, about 1.1 points per layer; Table 9), and even a one-layer model, which consumes the neighbourhood mean exactly as the linear model does, stays 3.7 points below neighbourhood-augmented LR.

The neighbourhood representation is the less stable one. How far each stage's centroid moves between training and test pucks, in units of its own within-class spread, separates the two feature spaces (Fig. 1A). Both are nearly stationary within a species (0.12 own, 0.19 neighbourhood) and under the *ob/ob* condition (0.09, 0.16), and both move across species (0.35, 0.46). In all 24 stage-by-puck comparisons the neighbourhood centroid moves further than the bead's own, by factors of 1.06 to 3.30. Round spermatids show both the largest cross-species neighbourhood displacement (0.53) and the largest GraphSAGE loss (Fig. 1B,C), which with four stages we read as consistent with the mechanism rather than evidence for it.

Secondary observations. The gain runs opposite to the target puck's neighbourhood homogeneity ($r = -0.52$, 95% CI $[-0.76, -0.12]$): the largest gains fall on the human pucks, which are the least homogeneous. It tracks instead how informative the neighbourhood is relative to the bead ($r = 0.69$, CI $[0.37, 0.88]$), a within-sample association rather than a usable predictor since both come from the same labels. Smoothing peaks at $\alpha = 0.5$ everywhere and the gain concentrates in low-UMI beads (appendix).

5 Discussion

The gains are small because the neighbourhood is largely redundant with the bead: a neighbourhood-only classifier recovers most of the signal the bead already carries. We would not expect this on niche-level labels, where the neighbourhood defines the class. Organization changes that matter at the scale of tubule architecture can be invisible at the scale of stage annotation, which is what the *ob/ob* comparison shows.

The more useful result is the split between how context is used. Fixed aggregation and post-hoc smoothing keep their gains under both shifts; a two-layer GNN matches them within a species and loses about five points across species. Averaged over all folds the GNN is not separable from expression only (9/15 positive, $p = 0.61$): its in-distribution gain and cross-species loss cancel. Both it and the linear model see the same neighbourhood mean, but only the GNN learns a function of it, and that is the less stable of the two representations. For reference-based annotation across species, a fixed spatial operator has an advantage that in-distribution benchmarks do not reveal.

Limitations. Labels are NMFreg assignments, not ground truth, and the human ones are noisier (in-distribution ceiling 0.49 vs. 0.70), so results measure agreement with a published annotation; the human estimates rest on two pucks. We varied GraphSAGE depth but not architecture family, so attention-based and heterophily-robust designs are untested, as are paired histology and cross-species alignment [Rosen et al., 2024, Lotfollahi et al., 2022]. Per-puck standardization touches the test puck; it is unsupervised, but may attenuate the shift we measure. Code and splits will be released.

References

Richard M Baldarelli, Cynthia L Smith, Martin Ringwald, Joel E Richardson, Carol J Bult, and Mouse Genome Informatics Group. Mouse Genome Informatics: an integrated knowledgebase system for the laboratory mouse. *Genetics*, 227(1):iyae031, 2024. doi: 10.1093/genetics/iyae031.

Arjun Chakravorty, Jiwoong Yun, Henry Amrhein, Kevin L Colón, Takuya Sato, Benjamin D Simons, Shosei Yoshida, and Long Cai. The temporal architecture of the seminiferous epithelial cycle revealed by spatial transcriptomics. *Cell*, 189(13):4122–4139, 2026. doi: 10.1016/j.cell.2026.04.036.

Haiqi Chen, Evan Murray, Anisha Sinha, Anisha Laumas, Jilong Li, Daniel Lesman, Xichen Nie, James Hotaling, Jingtao Guo, Bradley R Cairns, Evan Z Macosko, C Yan Cheng, and Fei Chen. Dissecting mammalian spermatogenesis using spatial transcriptomics. *Cell Reports*, 37(5):109915, 2021. doi: 10.1016/j.celrep.2021.109915.

Kangning Dong and Shihua Zhang. Deciphering spatial domains from spatially resolved transcriptomics with an adaptive graph attention auto-encoder. *Nature Communications*, 13(1):1739, 2022. doi: 10.1038/s41467-022-29439-6.

Will Hamilton, Zhitao Ying, and Jure Leskovec. Inductive representation learning on large graphs. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

Rex A Hess and Luiz Renato de Franca. Spermatogenesis and cycle of the seminiferous epithelium. *Advances in Experimental Medicine and Biology*, 636:1–15, 2008. doi: 10.1007/978-0-387-09597-4_1.

Jian Hu, Xiangjie Li, Kyle Coleman, Amelia Schroeder, Nan Ma, David J Irwin, Edward B Lee, Russell T Shinohara, and Mingyao Li. SpaGCN: Integrating gene expression, spatial location and histology to identify spatial domains and spatially variable genes by graph convolutional network. *Nature Methods*, 18(11):1342–1351, 2021. doi: 10.1038/s41592-021-01255-8.

Qian Huang, Horace He, Abhay Singh, Ser-Nam Lim, and Austin R Benson. Combining label propagation and simple models out-performs graph neural networks. In *International Conference on Learning Representations*, 2021.

Bin Li, Wen Zhang, Chuang Guo, Hao Xu, Longfei Li, Minghao Fang, Yinlei Hu, Xinye Zhang, Xinfeng Yao, Meifang Tang, et al. Benchmarking spatial and single-cell transcriptomics integration methods for transcript distribution prediction and cell type deconvolution. *Nature Methods*, 19(6):662–670, 2022. doi: 10.1038/s41592-022-01480-9.

Yahui Long, Kok Siong Ang, Mengwei Li, Kian Long Kelvin Chong, Raman Sethi, Chengwei Zhong, Hang Xu, Zhiwei Ong, Karishma Sachaphibulkij, Ao Chen, et al. Spatially informed clustering, integration, and deconvolution of spatial transcriptomics with GraphST. *Nature Communications*, 14(1):1155, 2023. doi: 10.1038/s41467-023-36796-3.

Mohammad Lotfollahi, Mohsen Naghipourfar, Malte D Luecken, Matin Khajavi, Maren Büttner, Marco Wagenstetter, Žiga Avsec, Adam Gayoso, Nir Yosef, Marta Interlandi, et al. Mapping single-cell data to reference atlases by transfer learning. *Nature Biotechnology*, 40(1):121–130, 2022. doi: 10.1038/s41587-021-01001-7.

Samuel G Rodriques, Robert R Stickels, Aleksandrina Goeva, Carly A Martin, Evan Murray, Charles R Vanderburg, Joshua Welch, Linlin M Chen, Fei Chen, and Evan Z Macosko. Slide-seq: A scalable technology for measuring genome-wide expression at high spatial resolution. *Science*, 363(6434):1463–1467, 2019. doi: 10.1126/science.aaw1219.

Yanay Rosen, Maria Brbić, Yusuf Roohani, Kyle Swanson, Ziang Li, and Jure Leskovec. Toward universal cell embeddings: integrating single-cell RNA-seq datasets across species with SATURN. *Nature Methods*, 21(8):1492–1500, 2024. doi: 10.1038/s41592-024-02191-z.

Vipul Singhal, Nigel Chou, Joseph Lee, Yifei Yue, Jinyue Liu, Wan Kee Chock, Li Lin, Yun-Ching Chang, Erica Mei Ling Teo, Jonathan Aow, et al. BANKSY unifies cell typing and tissue domain segmentation for scalable spatial omics data analysis. *Nature Genetics*, 56(3):431–441, 2024. doi: 10.1038/s41588-024-01664-3.

Robert R Stickels, Evan Murray, Pawan Kumar, Jilong Li, Jamie L Marshall, Daniela J Di Bella, Paola Arlotta, Evan Z Macosko, and Fei Chen. Highly sensitive spatial transcriptomics at near-cellular resolution with Slide-seqV2. *Nature Biotechnology*, 39(3):313–319, 2021. doi: 10.1038/s41587-020-0739-1.

Tim Stuart, Andrew Butler, Paul Hoffman, Christoph Hafemeister, Efthymia Papalexi, William M Mauck, Yuhan Hao, Marlon Stoeckius, Peter Smibert, and Rahul Satija. Comprehensive integration of single-cell data. *Cell*, 177(7):1888–1902, 2019. doi: 10.1016/j.cell.2019.05.031.

Marco Varrone, Daniele Tavernari, Albert Santamaria-Martínez, Logan A Walsh, and Giovanni Ciriello. CellCharter reveals spatial cell niches associated with tissue remodeling and cell plasticity. *Nature Genetics*, 56(1):74–84, 2024. doi: 10.1038/s41588-023-01588-4.

F Alexander Wolf, Philipp Angerer, and Fabian J Theis. SCANPY: large-scale single-cell gene expression data analysis. *Genome Biology*, 19(1):15, 2018. doi: 10.1186/s13059-017-1382-0.

Zhiyuan Yuan, Fangyuan Zhao, Senlin Lin, Yu Zhao, Jianhua Yao, Yan Cui, Xiao-Yong Zhang, and Yi Zhao. Benchmarking spatial clustering methods with spatially resolved transcriptomics data. *Nature Methods*, 21(4):712–722, 2024. doi: 10.1038/s41592-024-02215-8.

A Additional results

Table 2: Puck statistics after filtering. Homogeneity is the mean fraction of a bead's k nearest spatial neighbours that share its NMFreg label, over all beads (k=5, 15, 30) and over germ-cell beads only (k=15).

Puck	Condition	Beads	Median UMI	Homog. k=5	k=15	k=30	Germ, k=15
WT1	WT mouse	29,178	719	0.524	0.474	0.432	0.538
WT2	WT mouse	29,607	782	0.524	0.470	0.426	0.533
WT3	WT mouse	39,181	604	0.561	0.507	0.464	0.561
DB1	*ob/ob* mouse	25,377	797	0.484	0.426	0.385	0.485
DB2	*ob/ob* mouse	35,629	426	0.537	0.477	0.433	0.530
DB3	*ob/ob* mouse	30,073	684	0.574	0.518	0.473	0.568
HU1	human	30,464	602	0.362	0.336	0.316	0.380
HU2	human	33,389	600	0.379	0.352	0.331	0.397

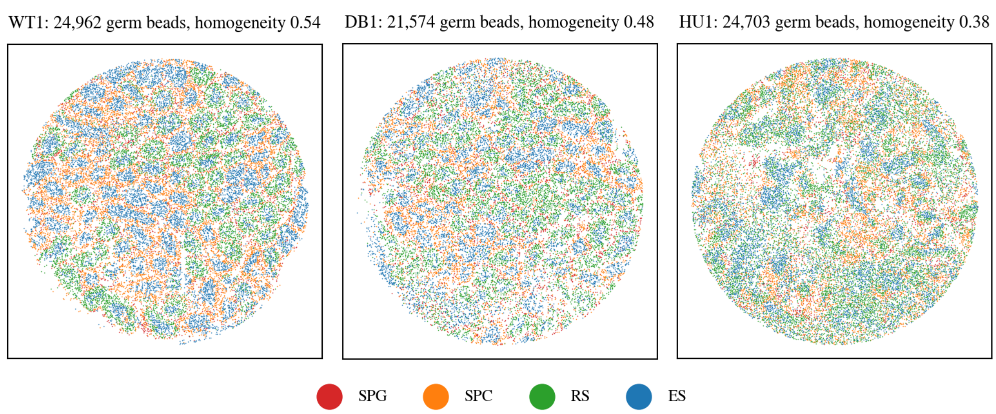

Figure 2: Germ-cell beads of one WT mouse, one *ob/ob* mouse, and one human puck, coloured by NMFreg stage. Tubule cross-sections with a spermatogonia/spermatocyte rim and a spermatid core are visible in mouse; human tubules are larger and stage labels are more mixed.

Table 3: Full version of Table 1, including the MLP models. Macro-F1, mean ± SD across test pucks.

Model	WT→WT	ob/ob→ob/ob	human→human	WT→ob/ob	WT→human	mouse→human
LR (expr)	0.699 ±0.013	0.717 ±0.022	0.486 ±0.008	0.713 ±0.026	0.385 ±0.005	0.385 ±0.006
LR (expr+nbr)	0.711 ±0.013	0.723 ±0.016	0.513 ±0.011	0.721 ±0.023	0.398 ±0.007	0.396 ±0.008
LR (nbr only)	0.636 ±0.003	0.607 ±0.019	0.464 ±0.006	0.609 ±0.029	0.365 ±0.012	0.355 ±0.008
LR (expr) + smooth (α=0.5)	0.712 ±0.016	0.733 ±0.019	0.501 ±0.008	0.732 ±0.021	0.401 ±0.005	0.400 ±0.006
MLP (expr)	0.693 ±0.014	0.704 ±0.036	0.475 ±0.012	0.704 ±0.028	0.370 ±0.009	0.369 ±0.009
MLP (expr+nbr)	0.705 ±0.018	0.708 ±0.029	0.498 ±0.014	0.713 ±0.023	0.378 ±0.013	0.375 ±0.011
GraphSAGE	0.711 ±0.015	0.716 ±0.023	0.505 ±0.013	0.721 ±0.023	0.347 ±0.012	0.350 ±0.012

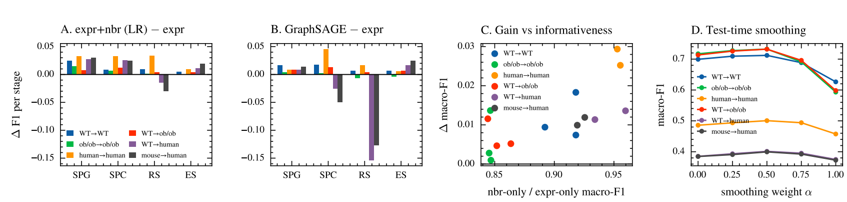

Figure 3: (A, B) Per-stage change in F1 from adding spatial context, relative to the expression-only LR, in all six settings; shared vertical scale. (C) Gain from neighbourhood features per test puck against the relative informativeness of context. (D) Test-time smoothing of the expression-only posterior against the weight α; $\alpha = 1$ replaces each bead's posterior by its neighbours' mean.

Neighbourhood size and smoothing weight. Table 4 reports the LR with neighbourhood features for $k \in \{5, 15, 30\}$ and test-time smoothing for all α.

Table 4: Sensitivity to neighbourhood size k (LR on expression plus neighbourhood mean) and to the smoothing weight α (expression-only LR posterior smoothed over the test puck's 15-nearest-neighbour graph). Macro-F1 for germ-cell stage, mean over test pucks.

	expr+nbr, k			smoothing weight α				
Setting	5	15	30	0	0.25	0.5	0.75	1
WT→WT	0.715	0.711	0.708	0.699	0.709	0.712	0.689	0.626
ob/ob→ob/ob	0.728	0.723	0.720	0.717	0.728	0.733	0.691	0.593
human→human	0.511	0.513	0.510	0.486	0.494	0.501	0.494	0.457
WT→ob/ob	0.722	0.721	0.718	0.713	0.726	0.732	0.696	0.598
WT→human	0.399	0.398	0.394	0.385	0.393	0.401	0.395	0.375
mouse→human	0.397	0.396	0.390	0.385	0.391	0.400	0.392	0.373

Six-class and nine-class labels. Tables 5 and 6 repeat the linear models on all beads with a six-class scheme (four germ stages, tubular somatic, interstitial) and with the original nine NMFreg labels. Macro-F1 is much lower for nine classes because the rare somatic types (endothelial, macrophage, myoid) are near chance from a single bead's expression even in-distribution. The gains from context are smaller than for germ-cell stage (0.4 to 1.0 points within a species for six classes, none for nine classes) but again largest on human in-distribution folds (about two points); across species with nine classes, neither neighbourhood features nor smoothing changes macro-F1 by more than 0.004.

Table 5: Six-class annotation (SPG, SPC, RS, ES, tubular somatic, interstitial), all beads. Macro-F1, mean ± SD across test pucks.

Model	WT→WT	ob/ob→ob/ob	human→human	WT→ob/ob	WT→human	mouse→human
LR (expr)	0.524 ±0.018	0.549 ±0.010	0.418 ±0.001	0.547 ±0.009	0.301 ±0.004	0.299 ±0.006
LR (expr+nbr)	0.533 ±0.017	0.553 ±0.007	0.441 ±0.002	0.553 ±0.007	0.308 ±0.003	0.306 ±0.002
LR (nbr only)	0.453 ±0.003	0.429 ±0.013	0.391 ±0.006	0.435 ±0.017	0.280 ±0.007	0.278 ±0.007
LR (expr) + smooth (α=0.5)	0.531 ±0.018	0.558 ±0.016	0.433 ±0.000	0.561 ±0.012	0.316 ±0.006	0.315 ±0.002

Table 6: Nine-class annotation with the original NMFreg labels, all beads. Macro-F1, mean ± SD across test pucks.

Model	WT→WT	ob/ob→ob/ob	human→human	WT→ob/ob	WT→human	mouse→human
LR (expr)	0.390 ±0.005	0.405 ±0.000	0.309 ±0.002	0.409 ±0.010	0.181 ±0.006	0.182 ±0.003
LR (expr+nbr)	0.392 ±0.004	0.401 ±0.004	0.330 ±0.001	0.413 ±0.012	0.177 ±0.005	0.180 ±0.004
LR (nbr only)	0.316 ±0.004	0.296 ±0.010	0.280 ±0.005	0.305 ±0.007	0.160 ±0.006	0.168 ±0.002
LR (expr) + smooth (α=0.5)	0.393 ±0.006	0.412 ±0.010	0.328 ±0.001	0.416 ±0.014	0.179 ±0.005	0.181 ±0.003

Gain by sequencing depth and local homogeneity. Table 7 stratifies the accuracy gain from neighbourhood features by the test bead's UMI tertile and by the fraction of its 15 neighbours that share its label. Within a species the gain comes from low-UMI beads; under species shift it comes from high-UMI beads. In every setting context lowers accuracy for beads with heterogeneous neighbourhoods (fewer than 40% same-label neighbours) and raises it for the rest, so the puck-level gain is a net of these two effects.

Table 7: Accuracy gain from neighbourhood features (expr+nbr minus expr, LR) stratified by the test bead's UMI tertile within its puck and by its local label homogeneity. Mean over test pucks of each setting.

	UMI tertile			local homogeneity				
Setting	low	mid	high	0-0.2	0.2-0.4	0.4-0.6	0.6-0.8	0.8-1
WT→WT	+0.023	+0.003	+0.003	-0.023	-0.010	+0.022	+0.030	+0.018
ob/ob→ob/ob	+0.015	-0.001	-0.002	-0.036	-0.011	+0.017	+0.024	+0.019
human→human	+0.047	+0.021	+0.018	-0.048	+0.014	+0.091	+0.111	+0.089
WT→ob/ob	+0.015	+0.002	+0.003	-0.035	-0.009	+0.019	+0.030	+0.021
WT→human	-0.011	+0.002	+0.029	+0.000	-0.007	+0.011	+0.034	+0.050
mouse→human	-0.001	+0.004	+0.016	-0.007	-0.001	+0.013	+0.033	+0.042

Representation shift. Table 8 reports both measures of how far the feature distribution moves between training and test pucks: the area under the receiver operating characteristic curve (AUC) of a logistic regression trained to separate training-domain from test-domain beads, and the per-stage centroid displacement of Fig. 1A. Displacement is normalised by the within-class spread in that same space, since averaging over 15 neighbours shrinks the spread as well as the offset; the AUC is not normalised and is reported only to establish that the *ob/ob* condition sits at chance in both spaces.

Table 8: Representation shift between training and test pucks. Domain-classifier AUC (0.5 = indistinguishable) and mean per-stage centroid displacement in units of within-class spread, for the bead's own embedding and for the neighbourhood mean.

		domain-classifier AUC		centroid shift	
Setting	Test	own	nbr	own	nbr
WT→WT	WT3	0.513	0.527	0.124	0.188
WT→ob/ob	DB1	0.499	0.534	0.067	0.148
WT→ob/ob	DB2	0.503	0.518	0.091	0.189
WT→ob/ob	DB3	0.495	0.504	0.116	0.141
WT→human	HU1	0.552	0.619	0.355	0.465
WT→human	HU2	0.551	0.620	0.342	0.448

Hyperparameters. All values were fixed before running the grid and shared across every setting; none were tuned on a test puck. Logistic regression: $C = 0.5$, class-balanced weights, at most 1000 iterations. MLP: 256 hidden units, dropout 0.2, AdamW with learning rate 10^{-3} and weight decay 10^{-4}, 40 epochs at batch size 512. GraphSAGE: two layers, 128 hidden units, dropout 0.2, AdamW with learning rate 3×10^{-3} and weight decay 10^{-4}, 150 full-batch epochs. PCA is fit on the training pucks only, with 50 components throughout.

GNN depth. Table 9 varies the number of GraphSAGE layers at otherwise identical settings. In distribution the three depths are within seed noise of each other; across species each additional layer costs about 1.1 macro-F1 points, and the ordering is the same on both human pucks. This is the prediction the mechanism in Fig. 1A makes: a one-layer model consumes the neighbourhood mean once, exactly as the linear model does, while deeper models learn a function of the neighbourhood distribution that does not hold in the target species.

Table 9: GraphSAGE macro-F1 by depth, germ-cell stage, mean ± SD over three seeds. All other hyperparameters are unchanged from the main results, in which GraphSAGE has two layers.

		GraphSAGE layers		
Setting	Test	1	2	3
WT→WT	WT1	0.697 $\pm$0.001	0.697 $\pm$0.001	0.695 $\pm$0.004
WT→human	HU1	0.355 $\pm$0.006	0.341 $\pm$0.006	0.331 $\pm$0.002
WT→human	HU2	0.366 $\pm$0.004	0.355 $\pm$0.003	0.347 $\pm$0.002

Model and neighbourhood-size sensitivity. Neighbourhood size changes macro-F1 by at most 0.008 across $k \in \{5, 15, 30\}$ (Table 4), and an MLP on the same features is 0.6 to 2.1 points below the logistic regression in every setting (Table 3), so neither the capacity of the classifier nor the radius of the neighbourhood drives the results.

Statistical tests. Across the 15 setting–test-puck folds, the gain from neighbourhood features over expression only is positive in all 15 (Wilcoxon signed-rank $p = 6 \times 10^{-5}$), as is the gain from test-time smoothing. A bead-level paired bootstrap within each test puck (400 resamples) gives a 95% interval excluding zero on 13 of 15 folds for neighbourhood features and 14 of 15 for smoothing. GraphSAGE is the exception: averaged over all folds it is not separable from the expression-only model (9 of 15 folds positive, $p = 0.61$), because its in-distribution gain and its cross-species loss cancel. Restricted to the four human-target folds, GraphSAGE is below neighbourhood-augmented LR by 0.048 on average, roughly twelve times the largest seed-to-seed standard deviation we observed for it. Correlations across the 15 folds, with 95% intervals from 5,000 bootstrap resamples of the folds, are $r = +0.69$ $[+0.37, +0.88]$ for relative informativeness against the gain and $r = -0.52$ $[-0.76, -0.12]$ for neighbourhood homogeneity; the sign of the second is the opposite of what the organization-based intuition predicts.